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(54) **WRITING INSTRUMENT**

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ABSTRACT

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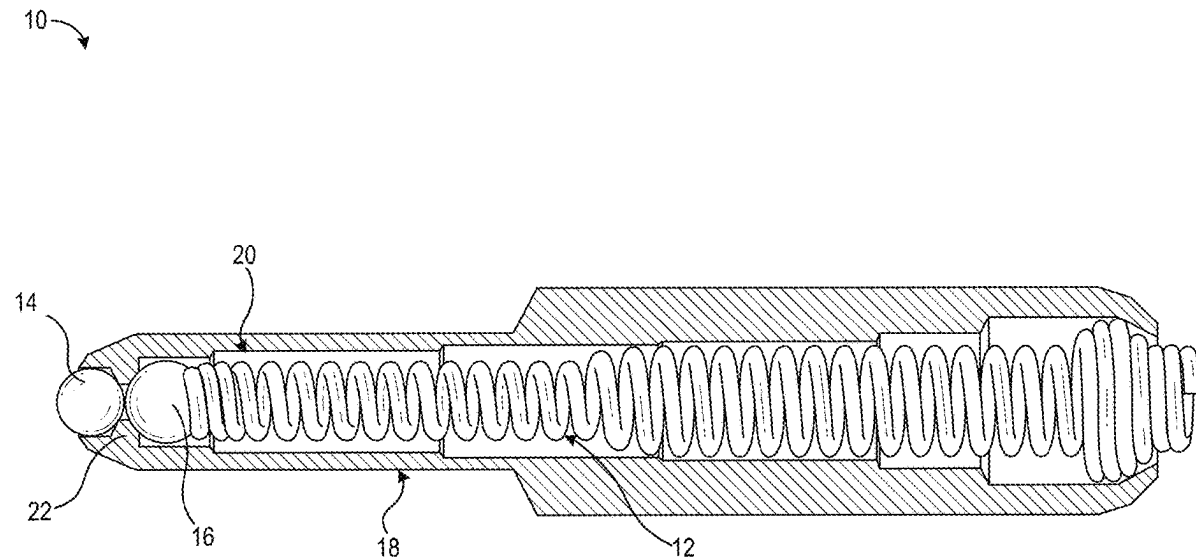
Related U.S. Application Data

(63) Continuation of application No. 17/663,655, filed on May 16, 2022, now Pat. No. 12,311,694.

A writing tip assembly for dispensing a free-flowing low-viscosity ink from a writing instrument may include a micro-spring, a writing ball configured to transfer ink to a substrate, and a second ball between the writing ball and the micro-spring. The second ball is configured to transfer ink to the writing ball. A writing instrument for dispensing a free-flowing low-viscosity ink includes an ink reservoir, the writing tip assembly, and a feeder fluidically coupling the ink reservoir to the writing tip assembly.

(30) **Foreign Application Priority Data**

May 19, 2021 (IN) 202111022407



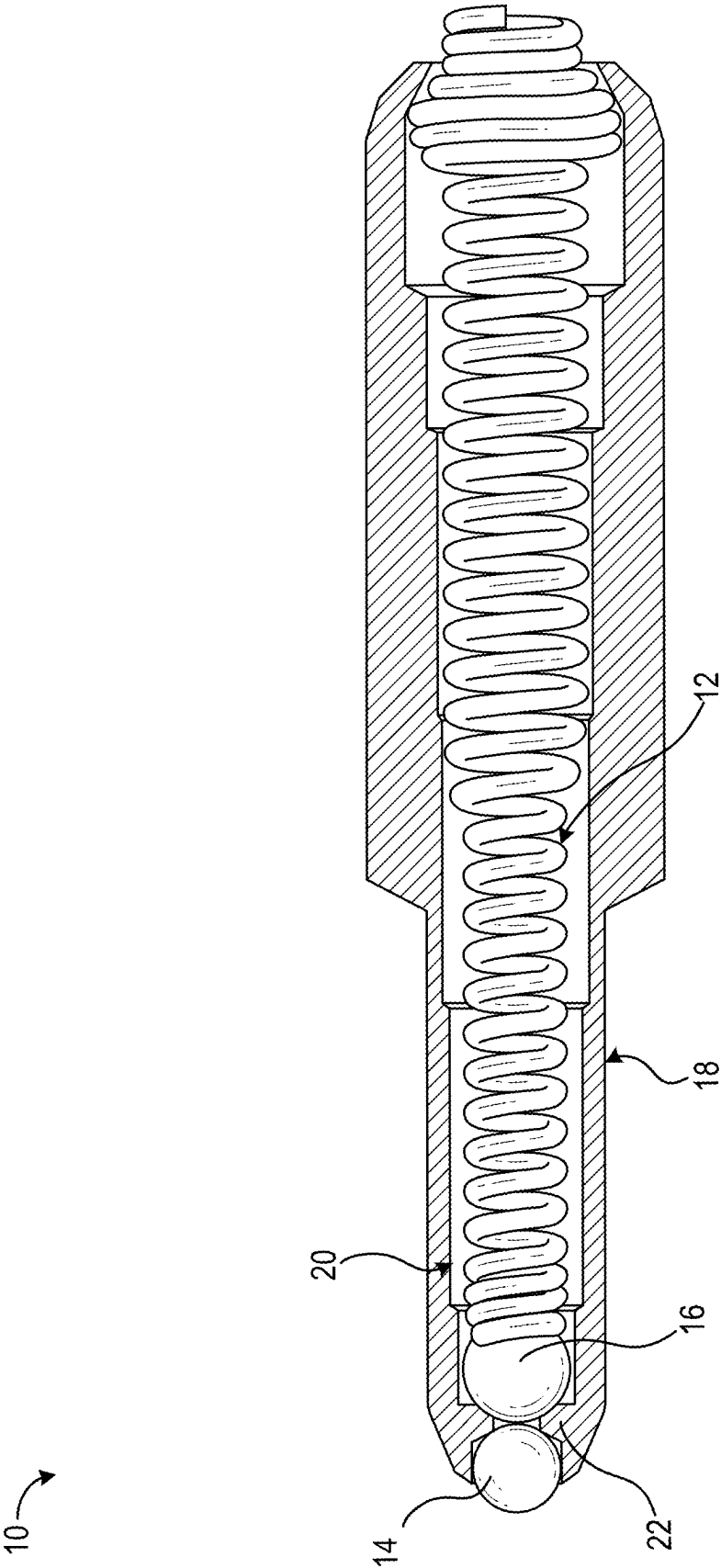


FIG. 1

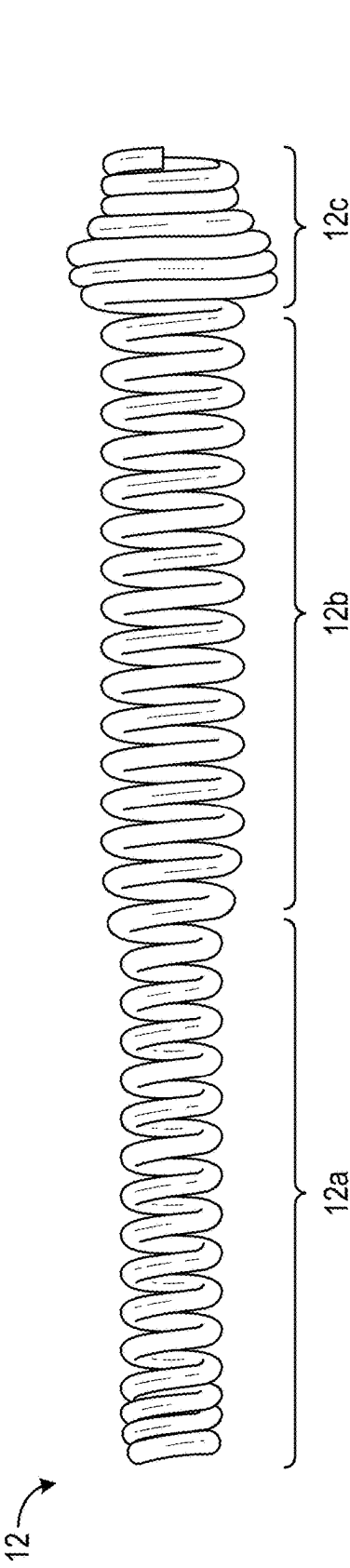


FIG. 2

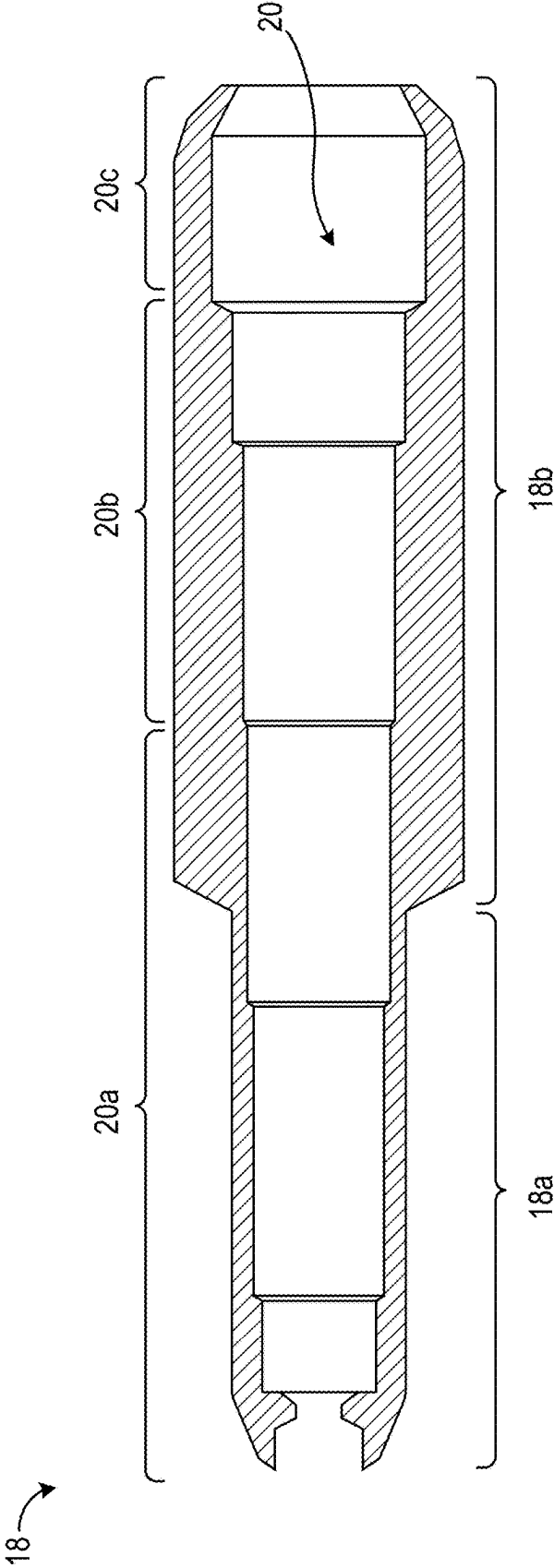


FIG. 3

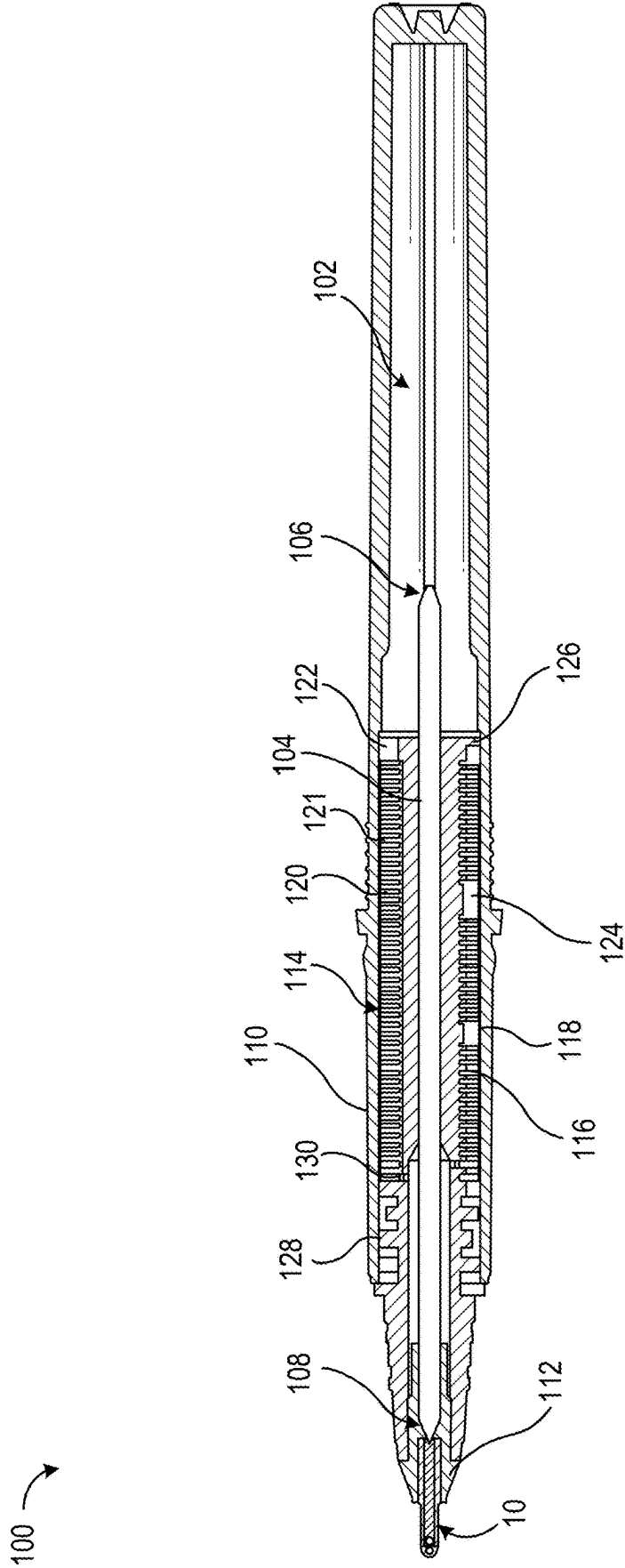


FIG. 4A

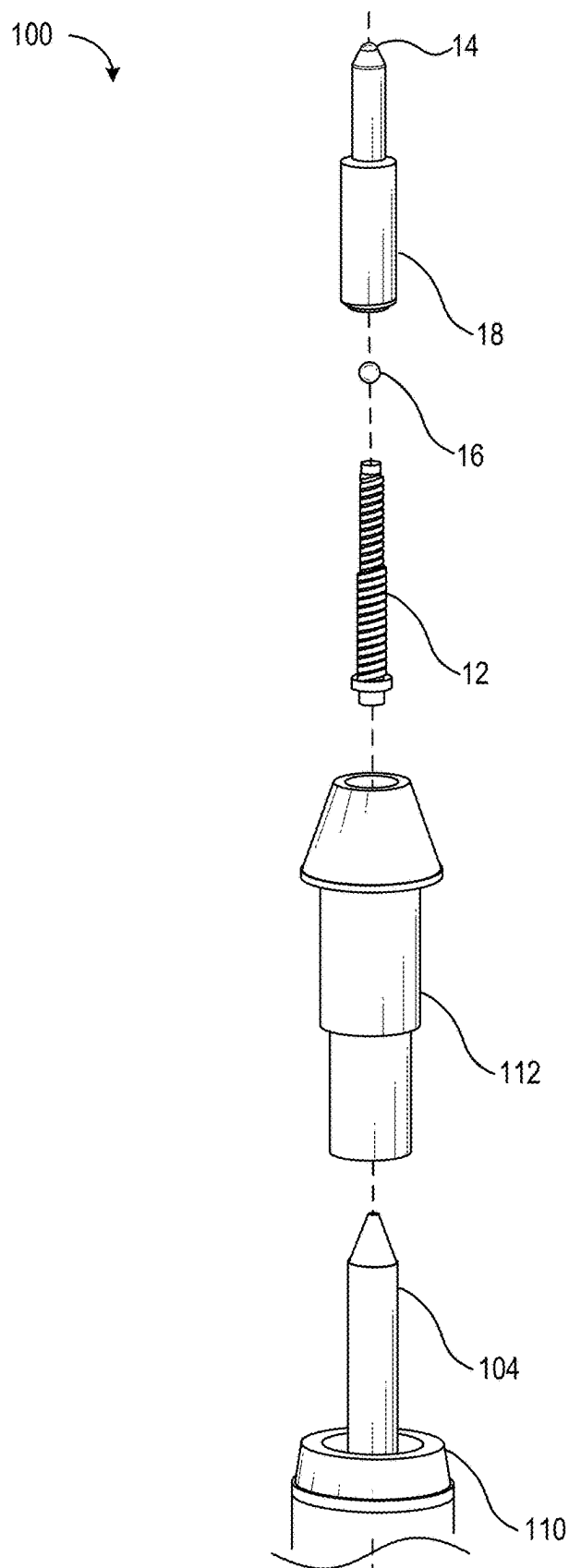


FIG. 4B

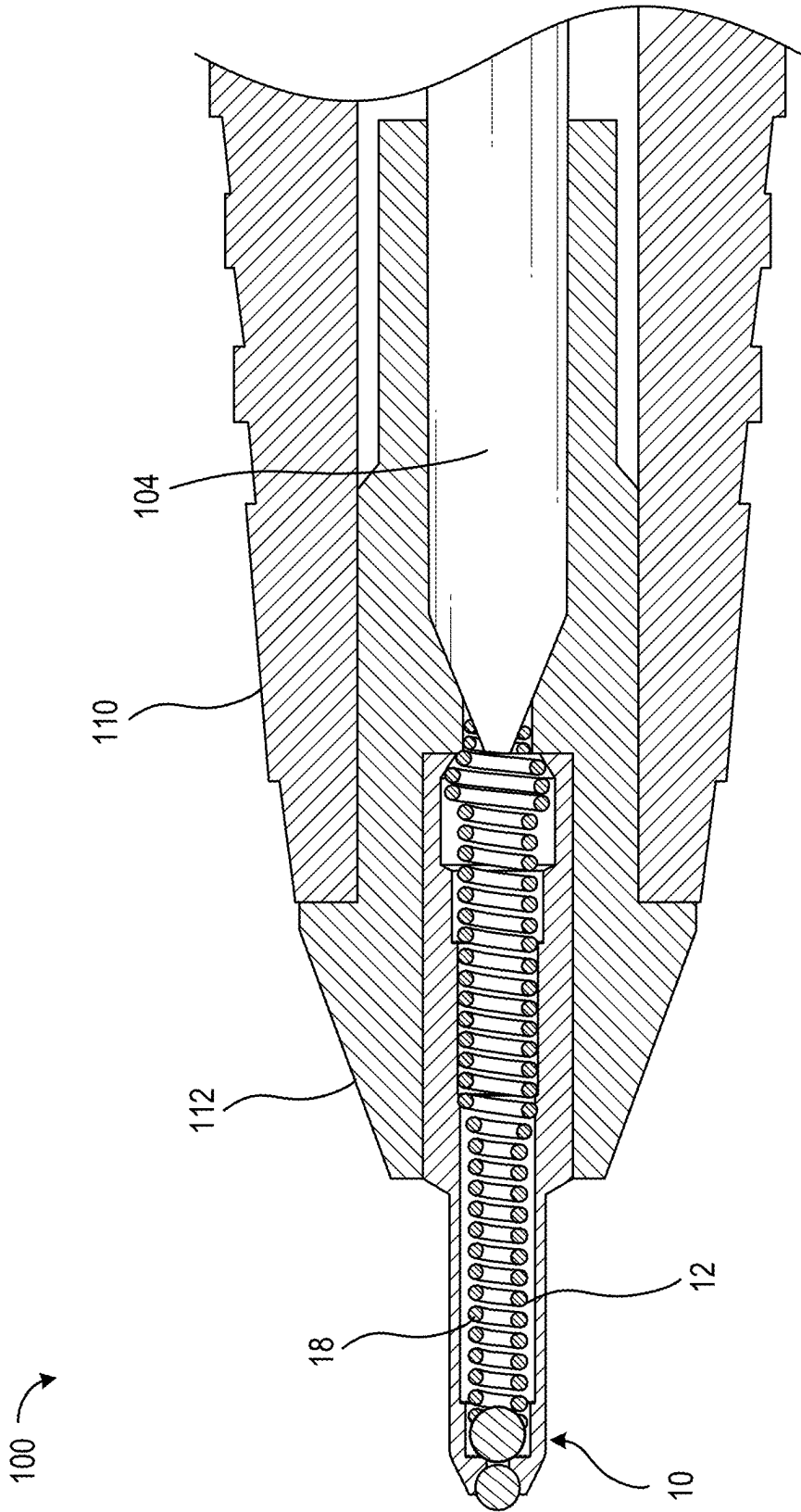


FIG. 4C

WRITING INSTRUMENT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 17/663,655, filed May 16, 2022, which claims priority to Indian Provisional Specification No. 202111022407, filed May 19, 2021, the contents of which are incorporated herein in their entirety by reference.

TECHNICAL FIELD

[0002] The present disclosure relates generally to writing instruments, for example, writing instruments including writing tip assemblies for dispensing free flowing ink.

BACKGROUND

[0003] One type of conventional writing instruments includes oil or gel-based ball-point pens, the viscosity of the ink used is around 300 to 10,000 mPa-s, which is relatively higher in order to avoid front tip leakage or back leakage issues. The high viscosity prevents leakage at different environmental conditions. To avoid leakage, the viscosity of the ink is maintained sufficiently high and thinner inks may be avoided. While gel-pens may include a spring associated with a writing tip, no feeder system or pressure regulation components are present. The only ink transfer member is the tip, with direct contact between the ink reservoir and the tip. The spring has a single coil diameter thorough its length except for the front and rear position. Such springs do not act as an ink transfer member, and may only control leaks at extreme environment conditions. The spring is not in contact with other elements other than the tip body.

[0004] Another type of conventional writing instruments use significantly thinner inks, for example, free flowing inks having low viscosities such as 1.5 to 60 mPa-s at standard temperature and pressure (STP) conditions. To control the discharge of the ink, a feeder is used, which includes a bundle of fibers packed together. The feeder allows the ink to be guided along a longitudinal direction to the tip from an ink reservoir. However, the free-flow system pen does not completely control the amount of ink flow/discharge. After certain meters of writing, the ink flow starts to diminish. This results in poor writing performance of the ink. The ink flow/discharge can be improved by weakening the capillary force. However, blow-out phenomenon may occur, where the ink leaks from the tip or through the lamella.

[0005] The blow-out phenomenon generally occurs when the outside pressure of the environment reduces. Other cases where the phenomenon may occur is with a decrease in ambient temperature, an increase in the pen temperature or an increase of the pressure within the free flow system pen. For example, this phenomenon may occur while travelling in aircraft such as planes. At greater altitudes, the atmospheric pressure drops. It drops to 0.3 bar at an altitude of 10000 m above the seal level at 15° C. and 0% Humidity. In such conditions, due the vast pressure difference present between the outside atmosphere and free flow system pen, the blow-out phenomenon occurs.

[0006] As per kinetic theory of gases, the viscosity and surface tension decrease in liquids when the temperature of the liquid rises. Thus, the tendency of blow-out phenomenon will be most likely in tropical or dry zones where the temperature varies and rises to an extremum of 50° C. At

such conditions, leakage may occur via the lamella/collector air inlet and tip point leakage.

[0007] A need remains for writing instruments that exhibit controlled dispensing of ink and with reduced, controlled, minimal, or substantially no leakage.

SUMMARY

[0008] The present disclosure describes writing tip assemblies and writing instruments including writing tip assemblies.

[0009] In aspects, the present disclosure describes a writing tip assembly for dispensing a free-flowing low-viscosity ink from a writing instrument. The writing tip assembly includes a micro-spring, a writing ball configured to transfer ink to a substrate, and a second ball between the writing ball and the micro-spring. The second ball is configured to transfer ink to the writing ball.

[0010] In aspects, the present disclosure describes a writing instrument for dispensing a free-flowing low-viscosity ink. The writing instrument includes an ink reservoir, the writing tip assembly, and a feeder fluidically coupling the ink reservoir to the writing tip assembly.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a conceptual cross-sectional view showing a writing tip assembly.

[0012] FIG. 2 is a conceptual side view showing a micro-spring of the writing tip assembly of FIG. 1.

[0013] FIG. 3 is a conceptual cross-sectional view showing a tip body of the writing tip assembly of FIG. 1.

[0014] FIG. 4A is a conceptual cross-sectional view of a writing instrument including the writing tip assembly of FIG. 1.

[0015] FIG. 4B is a conceptual partial exploded view of the writing instrument of FIG. 4A.

[0016] FIG. 4C is a conceptual partial cross-sectional view of the writing instrument of FIG. 4A.

DETAILED DESCRIPTION

[0017] The present disclosure describes writing tip assemblies and writing instruments including writing tip assemblies. Writing tip assemblies and writing instruments according to the disclosure may exhibit reduced or substantially no leakage under typical ambient conditions. Writing tip assemblies and writing instruments according to the disclosure may also provide relatively smooth writing by promoting uniform flow of ink.

[0018] Leakage through lamella air inlet may be termed as lamella leak, and may be controlled by providing enough space for the ink to occupy in the ink reserve groove and by increasing the number of circumferential fins. Leakage through the tip point may be controlled conventionally by reducing the ball rotation area, otherwise termed as ball play.

[0019] Conventionally, the ink is directly transferred to the writing tip or ball by a feeder or by any means, using capillary action. If the ball play is higher, the ink transfer will also be higher and vice versa. However, increasing the gap or ball play increases the risk of leakage. The lesser the ball play, lesser the ink transfer from the tip to the paper. Conventional ballpoint free flow pens may not be able to sufficiently control leakage via the tip writing ball when subjected to hot and humid environment conditions.

[0020] Blow-out phenomenon may eventually occur at very low pressures in conventional ballpoint free flow pens from the writing ball, because of the lesser resistance offered by the ink, where the force exerted by the viscosity of the fluid is overcome by the pressure differential. Thus, conventional free flow pens may exhibit tip point leakage irrespective of the pressure regulation performed by the collector/lamella inside the ink reservoir.

[0021] Thus, in free-flow system ink pens, water-based ink may be preferred because of its low viscosity and good surface tension characteristics. It ensures continuous feeding and proper ink discharge through the feeder. However, lower viscous ink may not provide good lubrication and may be susceptible to leakage. The free flow system utilizes the concept of capillary action, where the ink flows to the tip via the feeder (porous body with directional fibers). In conventional free flow systems, ink may be transferred to the tip either by direct contact with the surface of a feeder, or with the assistance of a plastic broach.

[0022] In aspects, the present disclosure describes a writing tip assembly for dispensing a free-flowing low-viscosity ink from a writing instrument. The writing tip assembly includes a micro-spring, a writing ball configured to transfer ink to a substrate, and a second ball between the writing ball and the micro-spring. The second ball is configured to transfer ink to the writing ball.

[0023] In aspects, the present disclosure describes a writing instrument for dispensing a free-flowing low-viscosity ink. The writing instrument includes an ink reservoir, the writing tip assembly, and a feeder fluidically coupling the ink reservoir to the writing tip assembly.

[0024] Writing instruments according to the present disclosure may use a low-viscosity ink and a free-flow system. Writing tip assemblies may include a double-ball and a micro-spring to promote ink flow and control while reducing or preventing leakage. The ink discharge may be controlled by both the double ball micro-spring tip and a lamella (or collector). Pressure regulation may be performed by the lamella. A feeder may be used to transfer ink to the tip assembly via capillary action of the ink. The micro-spring also may act as an ink-guiding member due to the negligible or very low pitch between coils of the micro-spring in the assembled tip. In some aspects, writing instruments according to the present disclosure use indirect ink transfer type—with no direct contact of the tip assembly with the ink reservoir. The micro-spring may contact the feeder to receive ink and enhance the fluid flow characteristics.

[0025] FIG. 1 is a conceptual cross-sectional view showing a writing tip assembly 10. The writing tip assembly 10 may be used for dispensing a free-flowing low-viscosity ink from a writing instrument (not shown in FIG. 1). The writing tip assembly 10 includes a micro-spring 12, a writing ball 14 configured to transfer ink to a substrate, and a second ball 16 between the writing ball 14 and the micro-spring 12. The second ball 16 is configured to transfer ink to the writing ball 14.

[0026] In aspects, the second ball 16 has a diameter greater than that of the writing ball 14. Providing such a greater diameter to the second ball 16 may promote retention of the second ball 16 against the writing ball 14 when biased by the micro-spring 12 and/or may promote a relatively greater transfer of ink to the writing ball 14 across the second ball 16.

[0027] FIG. 2 is a conceptual side view showing the micro-spring 12 of the writing tip assembly 10 of FIG. 1. The micro-spring 12, in addition to biasing the second ball 16 toward the writing ball 14, may also act as a flow medium for ink. For example, ink may flow along or within the interstices of the micro-spring 12, and a separate flow regulator along the length of the micro-spring 12 may not be necessary. In aspects, the micro-spring 12 includes a first spring portion 12a adjacent the second ball 16 and a second spring portion 12b extending away from the first spring portion 12a. The first spring portion 12a may have a smaller diameter than the second spring portion 12b. In some aspects, the first spring portion 12a has a smaller diameter than the second ball 16. In some aspects, the second spring portion 12b has a greater diameter than the second ball 16.

[0028] In aspects, the micro-spring 12 further includes a third spring portion 12c. The second spring portion 12b may be between the first spring portion 12a and the third spring portion 12c. The second spring portion 12b may have a smaller diameter than that of the third spring portion 12c. In some aspects, the third spring portion 12c has a smaller length than that of the second spring portion 12b. In aspects, providing succeeding narrow diameters to sections of the micro-spring 12 in a direction toward the writing ball may promote uniform flow of ink along the interstices of the micro-spring 12.

[0029] The micro-spring 12 with more than one spring portions may be unitary, for example, integrally formed. In some aspects, different spring portions may be welded to form the micro-spring 12.

[0030] In aspects, the writing tip assembly further includes a tip body 18 defining a lumen 20. The tip body 18 may be unitary, for example, integrally formed. The writing ball 14 may be confined to a space defined by the tip body 18. In aspects, about 70% of the writing ball 14 may be covered by the tip body 18, with a remaining portion protruding out from the tip body and available to contact a writing substrate.

[0031] FIG. 3 is a conceptual cross-sectional view showing the tip body 18 of the writing tip assembly of FIG. 1. In aspects, the tip body 18 includes a ball section 18a housing the writing ball 14, the second ball 16, and a tip portion of the micro-spring 12. For example, the tip portion of the micro-spring may be a portion of the first spring section 12a. In some aspects, the tip body 18 includes a spring section 18b extending away from the ball section 18a and housing a remaining portion of the micro-spring 12. In some such aspects, the spring section 18b has a maximum wall thickness greater than that of the ball section 18a. In some aspects, the spring section 18b has a maximum outer diameter greater than that of the ball section 18a. In some aspects, the micro-spring 12 is locked at an end of the tip body 18 by a technique known as “spinning.”

[0032] The end of the micro-spring 12 secured to the tip body 18 (for example, an end of the third spring section 12c) is locked by a bend of the material of the tip body 18 circumferentially inward toward the axis or center of the writing tip assembly 10. The last few coils, for example, the last two coils, of the micro-spring 12, which may have a smaller diameter than the rest of the micro-spring 12 or of the third spring section 12c, protrudes out of the tip body 18, to provide surface contact with a feeder, as described with reference to FIGS. 4A to 4C.

[0033] The micro-spring 12, in compression, tends to push the second ball 16 to move toward the writing ball 14, which in turn tends to push the writing ball 14. In this way, the writing ball 14 seals the writing end of the tip body 18. For example, the surface of the writing ball 14 makes a seal contact with the tip body 18, thereby acting as a non-return valve mechanism. The second ball 16 and the micro-spring 12 realigns the writing ball 14, maintaining the seal.

[0034] Additionally, better lubrication or smoothness in writing may be provided, by smoother ball. For example, when the writing pressure is higher, the second ball 16 gets loaded in the broach 22 area resulting with more friction, reducing the tendency of writing ball rotation. However, the combination of the micro-spring force, point-to-point contact, and lubrication behavior present in the low viscous fluid ultimately provides smoother writing. The parameters may be adjusted to improve writing smoothness.

[0035] Ink gets filled in a hollow portion of the tip body adjacent the writing ball 14. Then the ink flows via the broach 22 of the writing tip assembly 10, filling an ink well formed adjacent the writing ball 14. When the writing ball 14 is rotated/displaced, ink flows out of the writing tip assembly 10 and runs as the writing ball 14 guides.

[0036] In some aspects, the writing tip assembly 10 has a diameter of 1.60 mm, and length of 8 mm. The writing ball 14 may have a diameter of 0.50 mm, locked at the front end of the tip, but able to freely rotate upon writing. The inner diameter of the tip body 18 adjacent the second ball 16 may have a diameter of 0.60 mm. The second ball 16 may have a diameter of 0.60 mm. The micro-spring 12 may be of 8 mm free uncompressed length with varying diameters. The rear end of the micro-spring 12 may have a larger diameter of 1.16 mm next to a section of 0.72 mm. Another section of 0.72 mm coil diameter may protrude out of the tip body to a length of 0.25 mm from the rear end of the tip that makes surface contact with the feeder 104. The larger coil diameter of the spring 1.16 mm may be locked inside the tip by a method called "spinning."

[0037] Spinning is a method of locking the micro-spring 12 onto the tip body 18 by bending the material of the tip body 18 at an end to a diameter lesser than the maximum diameter of the micro-spring 12. In some aspects, spinning includes bending the material to an included angle of 100° and a bent inner diameter of 1.05 mm, which is 0.10 mm lesser than the maximum spring dimension. In some aspects, the number of coils in the micro-spring 12 is 25, in which upon compression the pitch of the spring becomes negligible and acts an ink-guiding member. Using capillary action phenomenon, the ink passes from the feeder to the writing ball with the aid of the micro-spring 12.

[0038] The lumen 20 may at least partially house the micro-spring 12. For example, an end or a portion of the micro-spring 12 may protrude or extend beyond the lumen 20. In the embodiment shown in FIG. 1, a portion of the third spring portion 12c extends beyond the lumen 20. In some aspects, the lumen 20 substantially completely houses the micro-spring 20.

[0039] The lumen 20 may completely house the second ball 16. In some aspects, the lumen 20 at least partially houses the writing ball 14. For example, a rolling portion of the writing ball 14 may extend out from the lumen 20 to contact a substrate to leave a mark or an impression by transferring ink to the substrate.

[0040] The lumen 20 may have a substantially constant inner diameter along an entire length of the lumen 20. In some aspects, the lumen 20 may include different sections of different inner diameters along the length of the lumen 20. In some aspects, the lumen 20 defines a first lumen section 20a, a second lumen section 20b having a maximum diameter greater than that of the first lumen section 20a, and a third lumen section 20c having a maximum diameter greater than that of the second lumen section 20b. Providing different diameters to different sections of the lumen 20 may accommodate and retain different sections of the micro-spring 12 that may have different diameters. For example, a section of the micro-spring 12 having a larger maximum diameter may be constrained or restricted from moving longitudinally into a section of the lumen 20 having a smaller maximum inner diameter. The respective diameters of the sections of the micro-spring 12 and/or the lumen 20 may vary along their respective lengths.

[0041] In aspects, the tip body 18 defines a broach 22 between the writing ball 14 and the second ball 16. The broach 22 may be integrally formed with or unitary with the tip body, or may be formed as a distinct element, for example, a collar or a gasket. In some aspects, the broach 22 may be star-shaped.

[0042] The second ball 16 may be held between the broach 22 and the micro-spring 12. The broach 22 may define an ink aperture 24 fluidically coupled to the lumen 20, or constituting a neck or a narrowed section of the lumen 20. Thus, the broach 22 may be disposed with the lumen 20, or toward or near a writing end of the lumen 20, or substantially at the writing end of the lumen 20. The broach 22 may be in the form of a collar about an inner surface of the lumen 20, which may help retain the second ball 16 within the lumen 20, by resisting or preventing a migration or movement of the second ball 16 beyond an end of the lumen 20. The micro-spring 12 may bias the second ball 16 against the broach 22, and ultimately, in sufficient proximity to or in contact with the writing ball 14 so that ink about the second ball may be transferred across the broach 22 to the writing ball 14.

[0043] One or more of the micro-spring 12, the writing ball 14, the second ball 16, or the tip body 18 may include, consist of, or consist essentially of, a metal, an alloy, a ceramic, a glass, a plastic, or any other suitable rigid material or combinations thereof.

[0044] Examples of writing instruments including the writing tip assembly 10 are described with reference to FIGS. 4A to 4C. However, the writing tip assembly 10 described with reference to FIGS. 1 to 3 may be used in any suitable writing instrument. Further, any suitable writing tip assembly according to the present disclosure may be used in writing instruments described with reference to FIGS. 4A to 4C.

[0045] FIG. 4A is a conceptual cross-sectional view of a writing instrument including the writing tip assembly 10 of FIG. 1. FIG. 4B is a conceptual partial exploded view of the writing instrument 100 of FIG. 4A. FIG. 4C is a conceptual partial cross-sectional view of the writing instrument 100 of FIG. 4A.

[0046] The writing instrument 100 may dispense a free-flowing low-viscosity ink. The writing instrument includes an ink reservoir 102 for containing the ink. The writing instrument 100 may further include a feeder 104 fluidically coupling the ink reservoir 102 to the writing tip assembly 10.

The feeder **104** may include a plurality of porous fibers (for example, packed together as a bundle), and act as ink transfer member and to control the ink flow between the ink reservoir **102** and the writing tip assembly **10**. A portion of the feeder **104** may be immersed into ink in the ink reservoir **102** to help in better transfer of ink into the feeder **104**. The feeder **104** may thus include a fibrous substrate to promote regulated or controlled flow of ink from the reservoir along the feeder **104** toward the writing tip assembly **10**. In aspects, a maximum diameter of the micro-spring **12** is less than that of the feeder **104**. Providing a lower maximum diameter to the micro-spring **12** than that of the feeder **104** may allow the micro-spring **12** to transfer ink uniformly from the feeder **104** to the writing ball **14**.

[0047] The feeder **104** makes a surface contact with the micro-spring **12**, for example, at the coil turns protruding out of the tip body **18**. Ink transfer occurs via surface contact of the micro-spring **12** and the feeder **104**. The coil pitch of the spring **12** may be selected such that the compressed gap between coils is relatively very low, for example sufficiently low to sustain a capillary flow of ink along interstices between the coils.

[0048] The feeder **104** may extend from a reservoir end **106** to a tip end **108**. The reservoir end **106** may be at least partially inserted or in contact with ink in the reservoir **102**. The tip end **108** may contact an end of the micro-spring **12**, or otherwise an end of the lumen **20**, so that ink from the reservoir **102** ultimately flows through the feeder **104**, and through the lumen **20** (for example, along the micro-spring **12**), toward the writing ball **14**. For example, interstices between adjacent coils of the micro-spring **12** may define a capillary flow path for ink to flow from the ink reservoir **102** through the feeder **104** and the tip assembly **10** to the writing ball **14**.

[0049] The writing instrument **100** includes an instrument body **110** that defines the reservoir **102**, and that houses or holds components of the writing instrument **100**. The instrument body **110** may include a plastic, a metal, an alloy, a ceramic, or combinations thereof. The instrument body **110** may include transparent or translucent sections, or be transparent or translucent in entirety.

[0050] In some aspects, the ink reservoir **102** is filled with about 1.5 g ink having a very low viscosity of 1.5 to 60 mPa-s at room temperature. The lamella **114**, the adaptor **112**, and the feeder **104** may be assembled with the writing tip assembly **10** having flow characteristics matched with the flow parameters of the low viscous ink used in the writing instrument **100**.

[0051] In some aspects, the writing instrument substantially no leakage at up to 0.5 bar of vacuum pressure, either at the lamella **114** or at the writing tip assembly **10**.

[0052] In aspects, the writing instrument **100** further includes an adaptor **112** holding the tip assembly **10** adjacent the feeder **104**. For example, the adaptor **112** may be secured at or at least partially within the instrument body **110** adjacent a writing end. The adaptor **112** may include one or more cylindrical portions extending within the instrument body **110**, and a conical portion extend outside the instrument body **110**.

[0053] In aspects, the writing instrument **100** further includes a lamella **114** extending from the writing tip assembly **10** into the ink reservoir **102**. The lamella **114** may include a plurality of ribs or fins **116** circumferentially

mounted over (or integrally extending from) a cylinder **118** in equal intervals of space between one another.

[0054] Ink flow and airflow are two flows that play a role in pressure regulation. The pressure of the ink inside the ink reservoir **102** must substantially be equal to the ambient pressure, to avoid leakage. The lamella **114** facilitates pressure regulation.

[0055] The ink inside the ink reservoir **102** flows through the feeder **104** by capillary action in which the feeder **104** transfers the ink to the writing tip assembly **10**. A portion of the ink reservoir **102** is disposed around the periphery of the lamella **114**. The lamella **114** defines a number of types of grooves. For example, ink reserve circumferential grooves **120** formed by the gaps between the fins **116** may store ink temporarily. Air exchange grooves **121** or an ink guide groove **122** may act as a guide for ink flow to the circumferential ink reserve grooves **120**. A long groove that may be comparatively wider than the ink guide groove **122** is the ink reserve groove or airflow groove **124**. Both the grooves **122** and **124** may communicate with one another with the help of the equally spaced circumferential grooves **120**.

[0056] When the pressure inside the ink reservoir **102** increases relative to the atmospheric pressure either by increasing the temperature of the ink or by reducing the pressure outside the pen, the ink inside the ink reservoir **102** is conducted with the help of air exchanging groove **122** into the circumferential grooves **120** by displacing the air out which is present in the circumferential grooves **120**. The ink is guided by the ink guide groove **121** into the circumferential groove to the ink reserve groove **124**.

[0057] The pressure inside the ink reservoir **102** may get reduced, either by writing during which ink is released via the feeder **104**, by increasing the pressure outside the pen, or by lowering the temperature of the ink. In this case, the air present outside the system flows into the ink reservoir **102** where the ink is present in order to obtain equilibrium condition. The air exchange groove **122** is a groove where the lamella **114** and the ink reservoir **102** can communicate with one another so that the pressure inside the ink reservoir **102** gets regulated.

[0058] Airflow occurs in the lamella **114** with the outside environment using an air inlet hole. For example, air flows along the grooves **124** present in between the fins **116** reaching the air flow groove **121**. Then the air flow groove **121** communicates with the air exchange grooves **122** via the circumferential fins **116** thereby regulating the pressure inside the ink reservoir **102**.

[0059] The air exchanging portion of the air exchange grooves **122** is the member made in order to exchange the air between the ink reservoir **102** and the lamella **114**. A larger fin **126** may act as a bottom seal member, having an interference fit with the ink reservoir **102** circumferentially, so that the ink stored in the ink reservoir **120** will not be able to leak or escape into the lamella **114**.

[0060] One or more forward fins **128** communicate the air supply with outer environment or atmosphere and create a seal along inner walls of the ink reservoir **102**, providing regulated airflow.

[0061] The lamella **114** defines a through-hole through which the feeder **104** extends. Two air holes **130** (for example, square holes) of a relatively small size may be defined in the lamella **114**, for communicating the airflow between the inside of the lamella **114** and the fins.

[0062] The air holes 130 maintain pressure equilibrium inside the lamella 114 to the circumferential fins 116, helping in preventing tip leakage due to the increase in pressure in the feeder 104 present inside the lamella 114. In this way, the lamella 114 may regulate pressure differences and provide a stable ink supply from the ink reservoir 102 to the writing tip assembly 10.

[0063] Writing tip assemblies and writing instruments according to the present disclosure may substantially resist or avoid general issues such as feathering, skip writing, goofing, and blobbing. For example, the micro-spring system provides continuous ink flow, constant discharge with the help of the feeder. The writing ball may act as a closing member to seal the flow area whenever writing pressure is removed, reducing or avoiding excess flow of the ink outside the tip assembly, which in turn avoids goofing. The continuous and constant discharge of the ink avoids feathering or skip writing issues.

[0064] Writing tip assemblies and writing instruments according to the present disclosure may provide advantages such as the following. (1) No tip leakage, even with low viscosity ink. The double ball spring-loaded assembly moves the ball toward the seat and create a seal at idle condition, which may control tip leak even at extreme environment conditions such as high temperature/humidity or low pressure. (2) Capillary action: the micro-spring inside the tip body makes surface contact with the feeder outside the tip serving as a path/guide for the ink and enhancing the ink flow using capillary action. (3) Smooth writing: the micro-spring double ball system provides better cushioning effect to the writing ball, providing very smooth writing feel and supports lubrication for the tip rotation in free flow system pen. (4) Writing parameters: feather, smudge and skip writing issues may be reduced or prevented, for example, completely avoided, with the aid of spring action in free flow system pen.

EXAMPLES

Example 1

[0065] Writing tests were performed using an auto-writing machine. A writing distance of about 1000 did not exhibit any writing issues, standard temperature and pressure (STP) conditions.

Example 2

[0066] Writing smoothness was measured. Writing smoothness is termed as the smoothness or effortless writing nature felt by the user of the pen. It can be measured in terms of the writing feel gauged by a user in the scale of 1 to 10. The higher number represents a poor or scratchy writing feel and the lower number represents smooth and effortless writing. Based on a survey taken with group of 100 peoples, a free-flow writing instrument with a micro-spring double-ball tip had a rating of 1 to 4, with the frequency of ratings 1 and 2 around 45 percent, which enumerates the smooth writing flow. The cushioning behavior provided by the micro-spring aids the lubrication and smooth writing feel of the free-flow writing instrument.

[0067] While the disclosure has been described with reference to a number of embodiments, it will be understood by those skilled in the art that the invention is not limited to such disclosed embodiments. Rather, the invention can be

modified to incorporate any number of variations, alterations, substitutions, or equivalent arrangements not described herein, but which are commensurate with the spirit and scope of the invention. Additionally, while various embodiments of the invention have been described, it is to be understood that aspects of the invention may include only some of the described embodiments.

1. (canceled)

2. A writing tip assembly comprising:

a spring;

a tip body defining a lumen at least partially housing the spring;

a writing ball configured to transfer ink to a substrate, wherein a majority of surface area of the writing ball is covered by the tip body; and

a second ball between the writing ball and the spring, wherein the second ball is configured to transfer ink to the writing ball;

wherein the spring comprises a first portion adjacent the second ball, a second portion extending from the first portion, a third portion extending from the second portion, and a fourth portion extending from the third portion;

wherein the second portion has a smaller diameter than that of the third portion, and the third portion has a greater diameter than that of the second portion; and wherein the writing instrument comprises a feeder configured to provide ink from an ink reservoir to the writing tip assembly via contact between the feeder and the fourth portion.

3. The writing tip assembly of claim 2, wherein the second ball has a diameter greater than that of the writing ball.

4. The writing tip assembly of claim 2, wherein the first portion has a smaller diameter than that of the second ball.

5. The writing tip assembly of claim 2, wherein the second portion has a greater diameter than that of the second ball.

6. The writing tip assembly of claim 2, wherein the writing tip assembly is configured to dispense a free-flowing low-viscosity ink from a writing instrument.

7. The writing tip assembly of claim 2, wherein the spring is unitary.

8. The writing tip assembly of claim 2, wherein the third portion has a smaller length than that of the second portion.

9. The writing tip assembly of claim 2, wherein the fourth portion has a diameter equal to that of the second portion.

10. The writing tip assembly of claim 2, wherein the lumen houses the second ball.

11. The writing tip assembly of claim 2, wherein the tip body defines a broach between the writing ball and the second ball, wherein the second ball is held between the broach and the spring, and wherein the broach defines an ink aperture fluidically coupled to the lumen.

12. The writing tip assembly of claim 2, wherein the lumen defines a first lumen section, a second lumen section having a maximum diameter greater than that of the first lumen section, and a third lumen section having a maximum diameter greater than that of the second lumen section.

13. The writing tip assembly of claim 2, wherein the tip body comprises a ball section housing the writing ball, the second ball, and a tip portion of the spring, wherein the tip body comprises a spring section extending from the ball section and housing a remaining portion of the spring, and wherein the spring section has a maximum wall thickness greater than that of the ball section.

14. The writing tip assembly of claim **13**, wherein the spring section has a maximum outer diameter greater than that of the ball section.

15. The writing tip assembly of claim **2**, wherein the tip body is unitary.

16. The writing tip assembly of claim **2**, wherein the tip body comprises a metal or an alloy.

17. The writing tip assembly of claim **2**, wherein one or more of the writing ball, the second ball, or the spring comprises a metal or an alloy.

18. A writing instrument for dispensing a free-flowing low-viscosity ink, the writing instrument comprising:

an ink reservoir;

a writing tip assembly comprising:

a tip body defining a lumen; and

a spring at least partially disposed within the lumen;

a lamella extending from the writing tip assembly into the ink reservoir, the lamella comprising:

a plurality of fins, wherein a larger fin is disposed at the bottom of the lamella such that the fin creates a bottom seal between the lamella and the ink reservoir; and

air holes configured to facilitate airflow between the inside of the lamella and the plurality of fins; and a feeder fluidically coupling the ink reservoir to the spring of the writing tip assembly;

wherein the feeder contacts a portion of the spring extends beyond the lumen; and

wherein a feeder end of the tip body is bent inward at least at a 100° angle to secure the spring within the tip body, such that a bent inner diameter of the tip body is less than a maximum diameter of the spring at the feeder end.

19. The writing instrument of claim **18**, wherein a maximum diameter of the spring is less than that of the feeder.

20. The writing instrument of claim **18**, wherein interstices between adjacent coils of the spring define a capillary flow path for ink to flow from the ink reservoir through the feeder and the tip assembly to the writing ball.

21. The writing instrument of claim **18**, further comprising an adapter holding the tip assembly adjacent the feeder.

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